Aerial Robotics CW2

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INTRODUCTION

Cooperative multi-agent UAV swarms can execute coordinated missions that exceed the capabilities of any individual drone, with applications spanning surveillance, logistics, environmental monitoring, and choreographed displays. Across these domains the core challenge is the same: how should multiple autonomous agents coordinate their motion to achieve a collective goal? This coursework investigates the design and evaluation of multi-UAV formation flight controllers, focusing on how a group of five drones can traverse a series of increasingly complex environments while maintaining geometric formation.

Problem Statement

Multi-UAV formation flight presents several coupled challenges: holding specific relative positions while the formation follows a commanded trajectory, which demands tight position control and compensation for disturbances; avoiding collisions with both environmental obstacles and other drones whose formation geometries bring them close by design; reconfiguring without collisions and recovering the intended geometry; and operating across environments that range from open space through static obstacle fields to dynamically moving threats, each of which imposes different demands on planning horizon and reactivity. Underlying all of these is a fundamental architectural question: whether coordination should be orchestrated by a single central planner with global knowledge, or emerge from local interactions between independent agents.

Objectives

This report evaluates two contrasting coordination paradigms applied to the same set of formation flight challenges. The centralised approach uses a multi-agent path planner that computes collision-free, speed-synchronised trajectories for all drones from a global world model. The decentralised approach uses artificial potential fields, in which each drone independently computes its motion from attractive and repulsive forces derived from local information. Both controllers are implemented within the ROS 2 and Aerostack2 framework and tested across four competition stages: changing formations on a circular orbit, traversal of narrow window slits, navigation through a static forest, and avoidance of dynamically moving obstacles. We compare completion time, collision avoidance, formation fidelity, and parameter sensitivity to characterise the trade-offs between global optimality and architectural simplicity.

METHODOLOGY

Centralised Approach

Our centralised implementation follows a multi-drone path planning strategy in which a single controller node has full knowledge of the environment, obstacle locations, and all drone states. Rather than drones making independent decisions, a central planner computes collision-free trajectories for every drone simultaneously, then dispatches these as waypoint sequences to each agent for execution.

The controller operates as a single ROS 2 node that interfaces with each drone through the Aerostack2 Python API. Each drone is addressed by namespace, and the controller issues high-level behaviour commands like takeoff, go_to or follow_path which Aerostack2 translates into low-level trajectory tracking. Because the planner holds the complete world model, it can reason globally about inter-drone separation, obstacle avoidance, and formation geometry before any drone moves.

Formations are defined as sets of 2-D offsets relative to a formation centroid. We implemented seven formation shapes (line, V-shape, square, diamond, orbit, grid, and staggered) each specified as a list of (dx, dy) pairs for five drones. Formation transitions are handled by smoothly interpolating each drone's offset vector between the old and new shape over a configurable number of timesteps.

At each planning timestep, every drone is assigned a target position derived from the centroid pose plus its oriented formation offset. In obstacle-free regions the planner commands these positions directly, meaning formation geometry is maintained by construction rather than by reactive control. Deviations only arise when the obstacle avoidance pipeline displaces a drone from its nominal offset, and in such cases the formation recovery blending step pulls the drone back toward its baseline position as soon as the local clearance permits.

Collision avoidance with the environment is handled by the radial push operation. Every waypoint that lies within the clearance zone $r_{\text{clear}} = r_{\text{obs}} + r_{\text{safety}}$ is projected outward onto the clearance boundary. Because the push is applied iteratively over multiple passes and at fine spatial resolution ($\sim$0.05 m), it guarantees that the final path maintains the required standoff distance from every obstacle.

When two drones' paths converge below a proximity threshold, the controller resolves the conflict through altitude separation rather than lateral displacement: the drone deviating more from its baseline is ramped upward by a fixed offset, smoothed over neighbouring timesteps.

Because obstacle avoidance lengthens some drones' paths relative to others, naive constant-speed execution would desynchronise the formation. The controller computes each drone's actual path length per execution chunk and scales its commanded speed proportionally so that all drones complete each chunk in approximately equal time, subject to bounds $[0.2,\ 1.5]$ m/s. The reference time is derived from the median baseline chunk distance at the nominal speed of 0.5 m/s.

Stage 4 extends the centralised planner to incorporate dynamic obstacle avoidance while preserving the pre-plan-and-execute architecture. To enable temporal awareness, we implemented a trajectory prediction model based on the scenario's specified initial positions and nominal speeds. By simulating linear extrapolation and elastic wall reflections forward from mission start, we generate a position estimate $p_k(t)$ for every obstacle. Because the drones move at a known base speed, the arrival time t_s at any waypoint s is deterministic, allowing the planner to query the exact predicted obstacle state without runtime estimation.

The collision resolution logic utilises an updated `push_path_clear` routine. If a drone's planned position at time t_s falls within the clearance ring of $p_k(t_s)$, the waypoint is projected outward to the ring's boundary. This iterative relaxation, along with pre-deflection and formation-recovery blending, ensures the final path maintains the required standoff across the entire traversal timeline.

Decentralised Approach

The decentralised architecture for Stages 1, 3, and 4 uses an iterative push-based planner paired with an artificial potential field (APF) reactive controller. Stage 2 uses a different formulation described in the Optimisation-Based Approach section below, since the narrow-passage geometry of that stage cannot be resolved by APF.

Artificial potential field reactive controller: Our decentralised implementation uses an artificial potential field strategy in which each drone independently computes its velocity command from a superposition of attractive and repulsive force vectors. No drone has knowledge of any other drone's planned path. The coordination emerges solely from local sensing of neighbour positions and obstacle proximity.

Unlike the centralised controller, there is no global planner. Each drone runs the same potential-field law at a fixed control rate of 10 Hz, reading its own position and the positions of neighbouring drones via the Aerostack2 interface. The only shared information is the mission definition (start point, end point, obstacle locations) loaded from the same `scenario1.yaml` configuration file.

A lightweight swarm coordinator advances the centroid at a constant commanded speed and broadcasts target positions, but it performs no path planning. Each drone decides its own motion independently based on the forces acting on it.

Potential field formulation.: At each control tick, the net force on drone i is

$$\mathbf{F}_i = \mathbf{F}_i^{\text{att}} + \sum_k \mathbf{F}_{ik}^{\text{obs}} + \sum_{j \neq i} \mathbf{F}_{ij}^{\text{drone}} + \mathbf{F}^{\text{ff}} \tag{1}$$

where the four terms are defined as follows.

Attraction pulls drone i toward its formation target $\mathbf{t}_i$ with a linear spring:

$$\mathbf{F}_i^{\text{att}} = K_{\text{att}}\,(\mathbf{t}_i - \mathbf{p}_i), \qquad K_{\text{att}} = 5.0 \tag{2}$$

Obstacle repulsion activates when drone i enters the clearance zone of obstacle k. Within this zone the force is a blend of radial and tangential components:

$$\mathbf{F}_{ik}^{\text{obs}} = \frac{K_{\text{obs}}}{d_{ik}^2}\Big[w\,\hat{\mathbf{r}}_{ik} + (1-w)\,\hat{\boldsymbol{\tau}}_{ik}\Big] \tag{3}$$

where d_{ik} is the distance to obstacle k, $\hat{\mathbf{r}}_{ik}$ is the unit radial vector away from the obstacle, $\hat{\boldsymbol{\tau}}_{ik}$ is the tangential direction aligned with the travel heading, and $w = (r_{\text{clear}} - d_{ik})/r_{\text{clear}}$ weights the blend so that drones near the obstacle boundary are steered tangentially around it while drones very close to the trunk receive a strong radial push.

Inter-drone repulsion prevents collisions between drones using an inverse-distance law that activates below a safety radius r_{safe}:

$$\mathbf{F}_{ij}^{\text{drone}} = K_{\text{drone}}\,\frac{1/d_{ij} - 1/r_{\text{safe}}}{d_{ij}^2}\,\hat{\mathbf{r}}_{ij}, \qquad K_{\text{drone}} = 0.4 \tag{4}$$

Feed-forward adds a constant velocity bias $\mathbf{F}^{\text{ff}}$ in the travel direction at the commanded traverse speed, ensuring forward progress even when attractive and repulsive forces partially cancel.

The net force is clamped to a maximum speed of 2.5 m/s and integrated over the timestep $\Delta t = 0.1$ s to produce the next waypoint, which is dispatched as a non-blocking `go_to` command.

Because the centroid advances at constant speed regardless of individual drone states, a drone that is deflected by an obstacle naturally experiences a growing displacement from its target and hence a proportionally larger restoring force. This provides emergent speed compensation without any explicit inter-drone communication about path lengths or velocities.

The decentralised approach does not explicitly command split or rejoin manoeuvres. When a drone encounters an obstacle, the repulsive field deflects it from its formation slot. Recovery is governed by the attraction force: once the drone clears the obstacle's influence zone, the attraction toward its formation target pulls it back. The rate of recovery depends on the gain K_{att} and the residual displacement. Because there is no global coordination of the deformation, different drones may split around the same obstacle on different sides. The tangential force component biases each drone to sweep in the direction aligned with forward travel, but this is determined independently per drone based on its own approach geometry.

Several design decisions shaped the controller. Tangential-radial blending was introduced after initial tests with purely radial repulsion caused drones to stall in front of obstacles. The tangential component ensures forward progress is maintained even at close range. Feed-forward velocity was added because, without it, the attraction force alone was insufficient to maintain the commanded traverse speed when repulsive forces opposed forward motion.

Traversing a Dynamic Environment: Stage 4 introduces five pillar obstacles moving at 0.5 m/s that bounce elastically off the field boundary. The reactive Artificial Potential Field (APF) used in earlier stages fails here for a structural reason: the repulsive force, evaluated from an obstacle's instantaneous position, can push a drone laterally into its path rather than away from it. This results in the drone moving directly in front of a dynamic body, making a collision inevitable. To resolve this, we extended the formulation along three axes to create a *Predictive APF* with temporal awareness.

To establish a reliable foundation for this approach, we first addressed the simulation's obstacle tracking pipeline. The provided `/dynamic_obstacles/locations` topic was found to publish stale, round-robin cached positions, which prevented accurate velocity estimation. We replaced this with the Gazebo physics engine's `/world/empty/dynamic_pose/info` topic, bridged via `ros_gz_bridge` and filtered by the `dynamic_obstacle_*` prefix. Velocities are obtained by differentiating consecutive observations and clamping them to the known nominal speed, effectively rejecting numerical jitter during wall bounces.

Building on this data, the repulsion evaluation was extended to use both current and predicted future positions of each obstacle. Given a lookahead Δt and estimated drone and obstacle velocities v_i and v_k, we define:

$$d_{\text{eff}} = \min\big(\|p_i - p_k\|,\ \|(p_i + v_i\Delta t) - (p_k + v_k\Delta t)\|\big)$$

The force direction is taken from whichever of the two gaps is smaller, capturing both immediate proximity and imminent closure without double-counting the stationary case. A closing-speed gain $g = 1 + K_c \max(0, -\dot{r}_{ik})$ further amplifies repulsion against obstacles on a head-on trajectory by up to $2.5\times$ the baseline. We utilized parameters $\Delta t = 0.3$ s and $K_c = 0.5$, tuned through empirical observation.

Furthermore, the drone formation strategy was relaxed to allow for more aggressive evasive manoeuvres, even if the formation must be broken temporarily. Since a rigid attraction gain can pull an evading drone back into an obstacle's path, we introduced a dynamic risk factor $R \in [0, 1]$ scaling with the deepest obstacle penetration. This factor is applied as:

$$K_{\text{att}}(R) = K_{\text{att},0}\,(1 - (1 - \alpha)R)$$

with $\alpha = 0.2$. When R approaches 1, the attraction gain is reduced by up to 80%, allowing the drone to prioritise evasion; as R returns to 0, the drone smoothly recovers its formation slot without explicit rejoin commands.

Finally, we retained a feed-forward brake mechanism but isolated it from drone-to-drone proximity to prevent nominal formation spacing from triggering deceleration. The velocity bias ramps linearly between 1.2 m (full feed-forward) and 0.5 m (zero feed-forward). This ensures the swarm's forward drive is only halted when an obstacle passes perpendicularly in front of its trajectory, providing a final layer of safety in dense dynamic environments.

Optimisation-Based Approach

Stage 2 presents a geometric constraint that the APF controller of Stages 1, 3, and 4 cannot resolve. The V-formation has a total width of 2.4 m and Window 2 provides a gap width of only 1.0 m, which requires the outer wings to displace 0.7 to 0.9 m laterally while advancing through the wall plane. Under the APF formulation, the wall repulsion exceeds the bounded attraction at close range and the outer wings stall near the gap edge. A gradient-descent model predictive control (MPC) formulation is therefore applied to Stage 2.

Cost Function Formulation: Based on the V-formation offsets, the wall line segments, and the desired clearance, the following cost function can be obtained (Equation 5). The planner optimises over the full flight path rather than resolving forces at each control tick.

$$J_i = w_{\text{wall}}\,\nu_i^2 + w_{\text{form}}\,\|\mathbf{p}_i - \mathbf{v}_i\|^2 + w_{\text{smooth}}\,\kappa_i^2 \qquad (5)$$

where ν_i is the clearance violation of drone i's path against the wall line segments, $\mathbf{v}_i$ is the V-formation slot of drone i, and κ_i is the discrete path curvature. The three weights are $w_{\text{wall}} = 120$, $w_{\text{form}} = 1.5$, and $w_{\text{smooth}} = 3.0$, selected to prioritise wall safety over formation fidelity while maintaining smooth trajectories. Gradient descent runs for 400 iterations at a learning rate of 0.03, with the per-iteration step clamped to 0.015 m. The required wall clearance is 0.35 m.

The wall-avoidance term alone cannot route the outer wings through a 1 m gap, because the V-slot targets for the outer wings lie outside the gap at Window 2. A soft-clamp operator is therefore applied to each V-slot target within 2 m of a wall, which smoothly pulls the x-coordinate of the target into the gap range with a margin of 0.30 m. This ensures that the formation-keeping term attracts each drone toward a physically reachable point. Once gradient descent converges, the optimised path is post-processed through 15 iterations of Laplacian smoothing combined with a wall push-out, followed by a monotonic-y pass that prevents backtracking.

Three-Segment Decomposition: In the centralised variant, a single planner jointly optimises all five drone trajectories. Because the two windows are offset in x, treating the traverse as a single optimisation would cause the planner to relax into a diagonal route that oscillates between the two gap x-positions. The mission is therefore decomposed into three segments: the approach through Window 1, the transition between windows, and the passage through Window 2 to the exit. Each segment is optimised independently and executed as a separate `follow_path` command. Formation compression emerges from the equilibrium between the wall-avoidance term and the formation-keeping term, which means no compression schedule is explicitly encoded. The altitude profile is derived analytically from the window heights through a smoothstep function.

In the decentralised variant, there is no global planner. Each drone independently runs the cost function in Equation 5, optimising only its own trajectory. The inputs per drone are the scenario YAML and its own formation index i, which

determines its V-offset. No drone accesses the state or plan of any other drone at any point. Coordination emerges from two shared invariants: identical environment knowledge, as all drones read the same YAML, and the shared V-formation definition. Because the planning inputs are deterministic, the five independent solvers produce paths that are geometrically consistent with each other without any explicit communication.

Experimental Results & Performance Evaluation

Stage 1

Both approaches share the same trajectory structure: the centroid follows a circular orbit of diameter 3.0 m. One full loop is divided equally among the scheduled formations, so that each formation occupies an equal arc segment. Within each segment, the first two-thirds of the arc is allocated to the formation transition (morphing from the previous shape to the current one) and the remaining third is a hold phase where the target formation is maintained without change. The first segment of each loop is a pure hold, since there is no preceding formation to blend from.

The centralised controller completed one full loop through all formations in 131.0 s, while the decentralised potential-field controller required 146.6 s. This disparity has several causes. The centralised planner pre-computes the entire circuit and dispatches waypoint chunks with per-drone speed synchronisation, meaning drones with longer paths during formation transitions are commanded at proportionally higher speeds. The decentralised controller, by contrast, must advance the centroid slowly enough that the attraction force can keep all drones converged on their moving targets. Increasing the centroid movement caused drones to lag behind their formation slots, and the resulting larger attraction forces drove drones into proximity conflicts with neighbours, particularly during transitions where formation offsets cross. The decentralised approach is therefore fundamentally limited in orbit speed by the interplay between convergence rate and collision safety.

Both controllers successfully completed all formation transitions without collisions. However, the mechanisms ensuring this differ substantially. The centralised planner enforces minimum inter-drone separation through an offset during the interpolation between formations. This is sufficient to compute collision-free paths without any altitude separation. The decentralised controller required fixed altitude stratification during every transition to achieve efficient and collision-free transitions. Without vertical separation, drones whose offset vectors crossed during interpolation would block each other from moving or even push other drones out of their formation position.

For the decentralised controller, the actual drone trajectories are shaped by the real-time force balance rather than a pre-planned path. This results in slightly less uniform transitions, particularly for formations where drones have to pass by each other closely.

Stage 2

Both approaches achieved collision-free traversal of all five drones through both windows, resulting in a 100% success rate for both variants, which confirms the reliability of the wall-avoidance formulation.

The centralised planner completed the mission in 125.8 s while the decentralised controller required 117.7 s. Offline verification yielded minimum wall clearances of 0.341 m and 0.350 m respectively, with identical minimum inter-drone spacing of 0.302 m in both cases. The total planning time was 2.95 s for the centralised joint optimisation and 1.96 s for the sum of the five independent decentralised planners.

The two methods produce qualitatively similar trajectories. This is a direct consequence of both variants sharing the same cost function (Equation 5), the same environment knowledge, and the same V-formation definition. The algorithmic difference lies in the scope of optimisation: the centralised formulation couples all five paths in one cost function so that inter-drone consistency is enforced explicitly, while the decentralised formulation solves five independent optimisations and inter-drone consistency emerges from formation determinism. The near-identical wall clearances and identical inter-drone spacing are a direct consequence of this shared formulation.

As shown in the centralised flight path plot (Figure 1), the planned trajectories smoothly compress the formation laterally at each window while restoring the full V-shape in open regions. The compression is stronger at W2 than at W1, which reflects the cost-function equilibrium adapting to the local gap width. The altitude profile confirms that drones ascend to the prescribed window heights of $z = 2.0$ m at W1 and $z = 3.75$ m at W2.

The decentralised paths (Figure 2) are qualitatively similar in structure with comparable safety margins, despite the absence of any inter-drone coordination during planning. This identifies the shared-YAML approach as a valid decentralised strategy when agents share environment knowledge and a common formation definition.

Narrow-Passage Failure of Reactive APF: The reactive APF controller of Stages 1, 3, and 4 was tested on the Stage 2 scenario in Gazebo. Across the parameter ranges that were explored, the controller was unable to route all five drones through Window 2 and the outer wings consistently stalled near the gap edge. When the attractive and repulsive forces are of comparable magnitude at close range, the drone enters a local force equilibrium and cannot progress. The optimisation-based formulation avoids this regime because it reasons about the entire path globally rather than about instantaneous force balance. This identifies MPC as the appropriate choice for narrow-passage scenarios where reactive APF is fundamentally limited.

Stage 3

Both the centralised and decentralised controllers successfully guided all five drones through the forest of seven obstacles without any collisions, yielding a 100% success rate for both approaches.

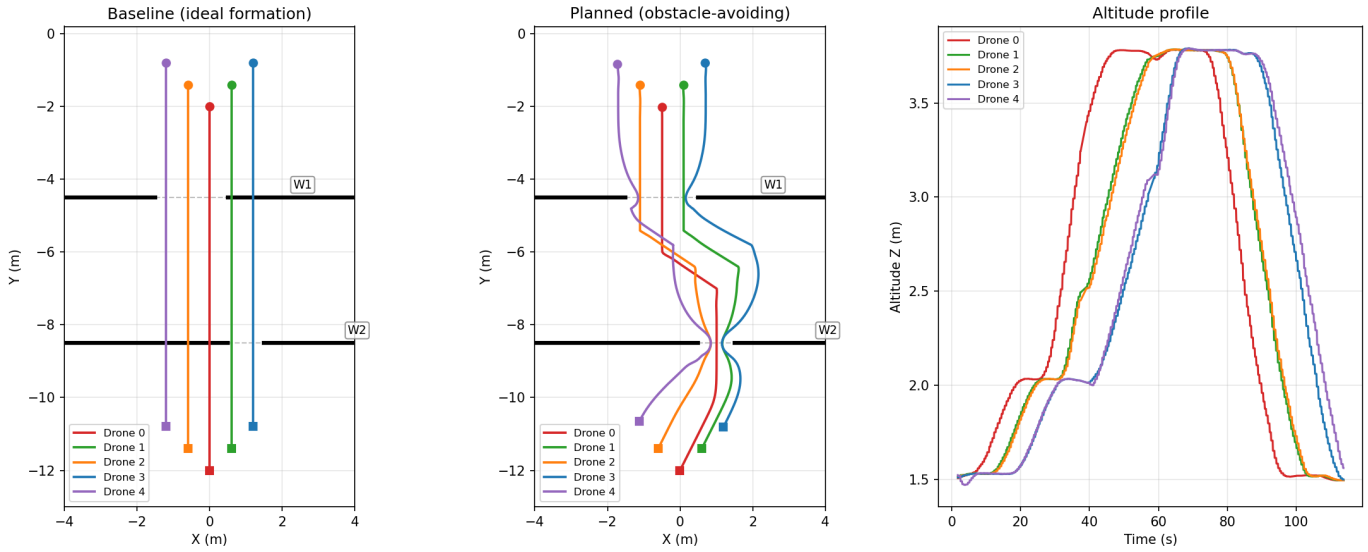


Fig. 1. Centralised approach for Stage 2. Showing ideal V-formation, planned obstacle-avoiding trajectories, and altitude profile throughout flight. Circle markers denote start positions; square markers denote end positions.

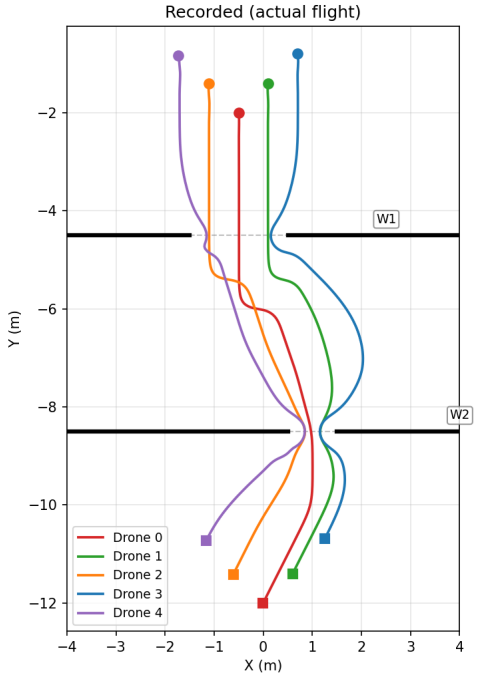


Fig. 2. Recorded flight paths for decentralised approach for Stage 2

With the centroid advancing at 1 m/s across the 8 m corridor, the centralised planner completed the traversal in 84.2 s while the decentralised controller required 126.2 s. This difference arises from how each approach handles obstacle detours. The centralised controller pre-computes all paths and applies per-drone speed compensation so that drones with longer detour paths fly faster, keeping the formation synchronised and completing each waypoint chunk in near-equal time. The decentralised controller, by contrast, has no explicit speed coordination. A drone deflected by an obstacle experiences a longer path at roughly the same force-limited velocity, and recovery depends entirely on the attraction force pulling it back toward its moving target. The additional time is therefore spent on post-detour catch-up rather than on the traverse itself.

As shown in the centralised flight path plot (Figure 3), the planned trajectories exhibit smooth, minimal deflections around obstacles, with the formation recovering to its baseline geometry as soon as clearance permits. The altitude profile confirms that where the flight paths of drone 2 and 4 converge laterally with other drone's flight paths near shared obstacles, the planner resolves the conflict vertically. It ramps one drone to 1.3 m while the other remains at 1.0 m avoiding any need for wider lateral displacement.

The decentralised paths (Figure 4) are qualitatively similar in that all drones maintain clearance from obstacles, but the deflections are less uniform. Because the obstacle repulsion force first starts to grow once a drone gets into the obstacle's safety margin, the drone only gets pushed around it once it's really close as only then the repulsive force becomes larger than the overall path attraction force.

The centralised approach maintains tighter formation coherence throughout the traverse. Speed compensation ensures all drones arrive at each waypoint chunk boundary simultaneously, so the line formation is preserved even through obstacle-dense regions. The only deviations from nominal spacing occur in the immediate vicinity of obstacles, and these are recovered within a few waypoint steps via the formation-recovery blending mechanism.

In the decentralised case, formation coherence degrades toward the end of the corridor. Drones that encountered obstacles along their path accumulate a positional lag relative to the centroid, which advances at constant speed regardless of individual drone positions. While the attraction force acts

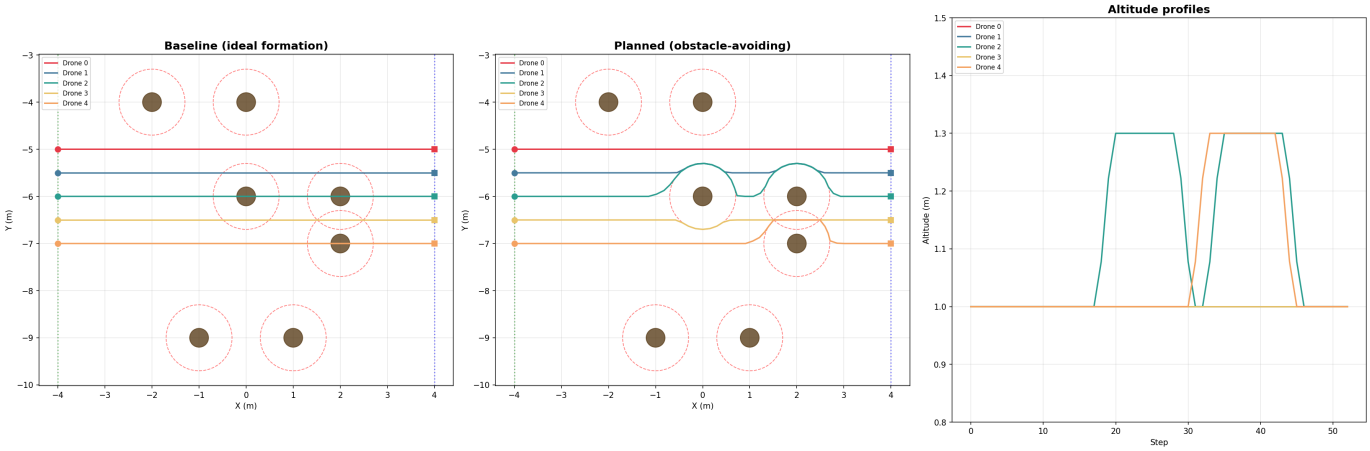


Fig. 3. Centralised approach for Stage 3. Showing ideal formation, planned formation (obstacle avoiding), and altitude profile throughout flight

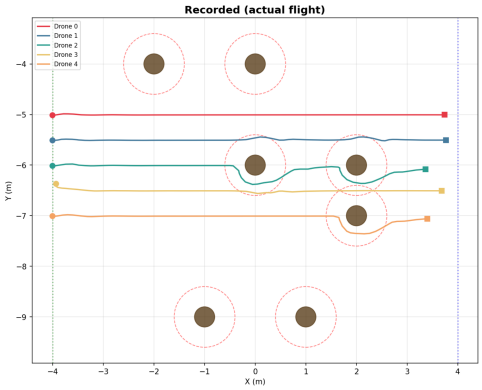


Fig. 4. Recorded flight paths for decentralised approach for Stage 3

as an implicit recovery mechanism, exerting a stronger pull on drones further from their target, the settling phase at the end of the traverse is insufficient to fully close the gap. The final formation error is modest (below 0.5 m) but visibly apparent in the recorded trajectories.

Both approaches achieve collision-free traversal, but the centralised planner offers superior time efficiency and formation fidelity due to its global path optimisation and explicit speed synchronisation. The decentralised controller trades these guarantees for architectural simplicity and the absence of a single point of failure, at the cost of longer completion time and reduced end-of-traverse formation accuracy.

Stage 4

The Predictive APF completed the 8 m traversal through the moving obstacle field without losing formation integrity. Over the 162 logged ticks of the traverse segment (representing 16.2 s of obstacle-field exposure within a 181.1 s total mission), the minimum drone-to-obstacle distance reached 0.106 m in the worst encounter. Specifically, 8 ticks registered contact below the 0.35 m obstacle surface, with 14 additional ticks falling within the near-miss band. The brake mechanism was active for 58 ticks, which correlates with the intervals where the hazard plot dips below the clearance threshold.

Figure X illustrates the swarm centroid and obstacle paths, colour-coded by time to allow direct interpretation of the temporal coincidence between drone positions and obstacle hits. Figure Y plots the minimum drone-to-obstacle distance over the traversal, featuring the 0.35 m obstacle surface and the 0.85 m clearance boundary for reference. Additionally, Figure Z isolates a representative manoeuvre where a single drone detours approximately 0.8 m from its formation slot to avoid an incoming pillar before successfully recovering its position. Throughout the mission, the formation centroid remained on the commanded straight-line trajectory, confirming that attraction suppression successfully relaxes individual slot pursuit without destabilising the swarm as a whole.

I. Discussion

Discussion

Table I summarises the completion time for both planning methods across all four stages.

TABLE I
COMPLETION TIME COMPARISON FOR BOTH PLANNING METHODS ACROSS ALL STAGES

	Stage 1	Stage 2	Stage 3	Stage 4
Centralised	131.0s	125.8s	84.2s	120.0s
Decentralised	146.6s	117.7s	126.2s	181.1s

For Stages 1, 3, and 4 the centralised approach is consistently faster, with the gap widening to roughly 50% in Stage 3 where obstacle detours benefit most from explicit speed synchronisation. Stage 2 is the only stage where the decentralised approach was marginally faster. Both variants use the same MPC formulation, so this gap is not algorithmic. The centralised variant dispatches three sequential `follow_path` commands (one per segment), each incurring a brief inter-segment handover delay, whereas the decentralised variant dispatches a single continuous path per drone. The gap therefore reflects execution overhead rather than planning performance.

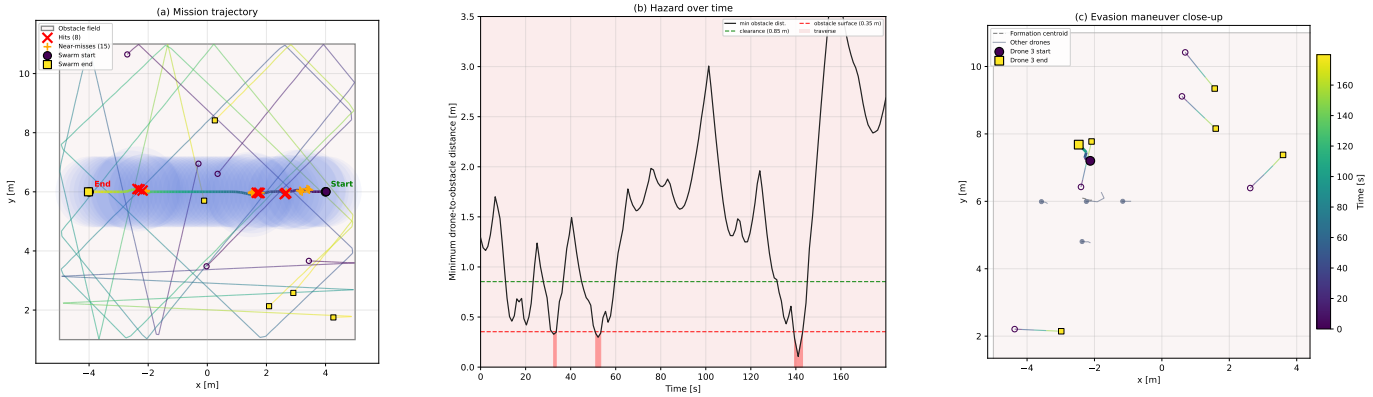


Fig. 5. Stage 4 Predictive APF results. (a) Mission trajectory with swarm centroid and obstacle paths colour-coded by time; red X markers indicate hits, orange + markers near-misses. (b) Minimum drone-to-obstacle distance over the traversal. (c) Close-up of a representative evasion manoeuvre, with the deviating drone in bold.

A fundamental difference in how the two approaches synchronise formation speed is worth highlighting. The centralised controller explicitly computes each drone's actual path length per chunk and scales its commanded speed accordingly so that all drones complete each segment in equal wall-clock time. The decentralised controller achieves the same effect emergently: a drone that has fallen behind experiences a larger attraction force and naturally accelerates toward its target. In Stage 3, drones that detoured around obstacles visibly trail the formation at the end of the traverse because the centroid advances at constant speed regardless of individual drone states, and the attraction force alone cannot fully close the gap within the available runway.

Stage 1 revealed a subtle failure mode unique to the decentralised approach. During formation transitions where offset vectors cross, one drone attempting to move through the gap between two others experiences strong repulsive forces from both neighbours, which can eject it from formation entirely rather than allowing passage. This was mitigated by fixed altitude stratification during every transition.

While the centralised planner produces cleaner and more predictable paths in Stage 3, we observed that the decentralised paths were in some cases shorter. The tangential-radial blended repulsion naturally finds efficient sweeping trajectories around obstacles, whereas the centralised push-based planner deflects paths to the clearance boundary somewhat conservatively. This does not translate to faster completion because the centralised speed compensates the offsets of longer paths but it is a notable property of the emergent approach.

Stage 4 exposed the fundamental performance ceiling of purely reactive local controllers. With drone speed bounded at 2.5 m/s and obstacle speed at 0.5 m/s, the worst-case closing speed is 3 m/s. Given our 0.5 m wide clearance ring, an obstacle can transit from first detection to surface contact in approximately 0.17 s—less than two control ticks at 10 Hz. Even with a perfectly tuned force field, this window is often too narrow to reliably arrest drone momentum through closed-loop position control. While the predictive extension effectively buys one extra tick of warning by projecting obstacle positions forward, it reduced rather than eliminated contacts. Closing this gap fundamentally would require a planner who reasons over the obstacle time-field, a direction we bypassed to maintain a decentralised architecture consistent with previous stages.

A secondary finding concerns the critical importance of data integrity. The most significant improvement in collision rates resulted not from control law tuning, but from replacing the provided obstacle topic with physics-engine ground truth. Prior to this change, lookahead, closing gain, and attraction suppression offered no measurable benefit because the controller operated on stale position data. This suggests that real-world deployment of such a system requires rigorous attention to the sensor pipeline; noisy or high-latency perception creates a bottleneck that cannot be compensated for solely at the control layer.

The centralised Stage 4 planner avoids the reactive ceiling by reasoning over the entire obstacle time-field before execution. Where the decentralised controller has at most two control ticks to react to an incoming pillar, the centralised planner routes around obstacles whose positions at the time of encounter are already part of the plan. This advantage is bounded by the accuracy of the obstacle motion model: our linear extrapolation with elastic wall bouncing is exact for the current scenario, but any unmodelled behaviour (collisions between obstacles, changes in speed, unknown trajectory shape) would require the planner to recompute mid-mission. The decentralised controller, by contrast, handles such cases for free because it only reasons about the present-tick configuration. This is the classic trade-off in swarm coordination: global optimality versus robustness to model error.

Challenges & Mitigations

Several challenges surfaced during development of the decentralised controller for the forest traversal. Initial versions using pure radial repulsion caused drones to stall in front of obstacles. The repulsive force directly opposed the attraction

force, producing a near-zero net force vector at the obstacle boundary and leaving drones deadlocked. We resolved this by introducing a tangential component blended with the radial push, biased to align with forward travel, so drones sweep around obstacles rather than pressing into them.

Stage 2 presented a challenge of a different nature: the reactive APF formulation used in the other stages could not resolve the narrow-passage geometry, as the outer wings of the V-formation consistently stalled at the gap edges where attraction and repulsion balance. Rather than attempting to tune APF gains around this local equilibrium, we switched to a gradient-descent MPC formulation that reasons over the entire flight path. Beyond the change of formulation, the dominant tuning challenge was parameter balance in the cost function, where the ratio of wall-avoidance weight to formation-fidelity weight determined whether the formation compressed cleanly through the gap or oscillated around the V-slot targets. A ratio of roughly 80:1 (w_{wall}=120, w_{form}=1.5) produced the tightest traversal while maintaining the required 0.35 m wall clearance.

Control rate selection for the decentralised controller also required experimentation. Early implementations ran the force computation at 20 Hz, which imposed unnecessary communication and computation overhead without improving flight quality. Reducing to 10 Hz improved overall performance with no observed degradation in stability or obstacle avoidance. Further reduction is possible in principle, but below roughly 5 Hz the reaction time becomes too slow for reliable obstacle avoidance at the traverse speeds used.

Finally, the parameter tuning effort was strikingly asymmetric between the two approaches. The decentralised controller's parameters interact non-trivially, and suitable values differed between Stage 1 and Stage 3 (for instance, the inter-drone safe distance needed to be widened from 0.15 m in Stage 3 to 0.27 m in Stage 1 to handle crossing manoeuvres during transitions). Each stage required its own tuning pass, largely by trial and error in simulation. The centralised controller's parameters (safety margin, clearance radius, chunk size, speed bounds) are more independent of one another and transferred between stages with minimal adjustment. This practical asymmetry should not be understated: despite the centralised planner's internal logic being more complex, it is substantially simpler to deploy and tune in practice than the conceptually simpler decentralised approach.

Future Improvements

Several directions could meaningfully improve the swarm coordination demonstrated in this work. A key limitation of the decentralised controller, observed particularly in Stage 1, is the difficulty of tuning the inter-drone repulsion force. The safe distance must be large enough that drones moving at high speed can decelerate before colliding, but this same range prevents drones from ever settling into tightly packed formations because they repel each other before reaching their target slots. A natural improvement would be to scale the repulsion force dynamically based on each drone's current velocity and its distance to its formation target: fast-moving drones receive strong early repulsion for safety, while drones approaching their designated slot at low speed see the repulsion decrease, allowing precise convergence.

Hybrid architectures combining the strengths of both paradigms offer perhaps the most promising direction. A two-layer controller could use centralised planning to compute globally optimal formation paths and speed schedules, while delegating reactive obstacle avoidance and fine inter-drone spacing to a decentralised potential-field layer running on each drone. This would retain the time efficiency and formation fidelity of the centralised approach while gaining the robustness of local reactive control, which is valuable when the world model is incomplete or when obstacles move unpredictably.

CONCLUSION

This work investigated centralised and decentralised coordination for a swarm of five drones across four scenario stages: formation changes on a circular orbit, narrow window traversal, static forest navigation, and dynamic obstacle avoidance. For the centralised approach we implemented a multi-agent path planning strategy with pre-computed trajectories, explicit speed synchronisation, and altitude separation. For the decentralised approach we used artificial potential fields with attraction, repulsion, and tangential force components, switching to an optimisation-based formulation for the narrow-passage Stage 2 where reactive forces proved fundamentally inadequate.

Both approaches achieved collision-free completion of all four stages, but with distinct performance characteristics. The centralised planner was consistently faster in the static scenarios, produced cleaner trajectories, and required substantially less parameter tuning. The decentralised controller demonstrated architectural simplicity, natural reactivity, and in some cases marginally faster completion (Stage 2), with emergent rather than hardcoded speed compensation arising from the attraction force itself. Stage 4 confirmed that predictive, planning-based methods generalise to dynamic environments through chunked replanning, while the purely reactive variant struggles with narrow passages regardless of scenario type.

Three broader findings emerge. First, the trade-off between global optimality and local autonomy is not absolute but scenario-dependent: centralised planning excels when the environment is static and fully known, while decentralised control offers advantages in reactivity and scalability that matter more as environments become dynamic or partially observed. Second, the algorithmic elegance of the decentralised approach is paid for in parameter tuning effort. Our potential-field gains required stage-specific adjustment, whereas the centralised parameters transferred between stages with minimal change. Third, the choice of coordination strategy interacts with the geometry of the task: Stage 2 revealed that reactive potential fields are fundamentally limited in narrow-passage scenarios due to local force equilibria, which motivated our switch to optimisation-based planning for that stage.